

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
Lecture: CPR 51

Title: Acid Base balance: Role of kidney

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Lecture: Acid Base Balance: Role of Kidney

SOM.MK.I.BPM1.3.CPR.1.INTG. 0009	Summarize and analyze the role of kidney in maintaining serum bicarbonate levels.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0010	Differentiate the processes of reabsorption of filtered HCO_3^- , formation of new HCO_3^- and secretion of HCO_3^- in renal tubule.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0011	Explain net acid excretion by the kidneys, titratable acid, and the production and excretion of ammonium.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0012	Explain the process of H^+ secretion in renal tubules by the carbonic anhydrase system. Identify factors affecting this process.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0013	Discriminate the role of phosphate and ammonium buffers in the renal tubule. Highlight the significance of glutaminase in prolonged acidosis.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0014	Given a sudden increase or decrease in pH, identify the magnitude and the time course of the compensations that act to minimize changes in pH of the body fluids, including: a) buffers, b) respiratory adjustments, and c) renal adjustments.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0015	Describe the interactions between K^+ and H^+ and how this could cause acid-base disturbances.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0016	Describe the effects of carbonic anhydrase inhibitors (acetazolamide) and the other diuretics on acid-base balance and the reabsorption of HCO_3^- by the nephron.

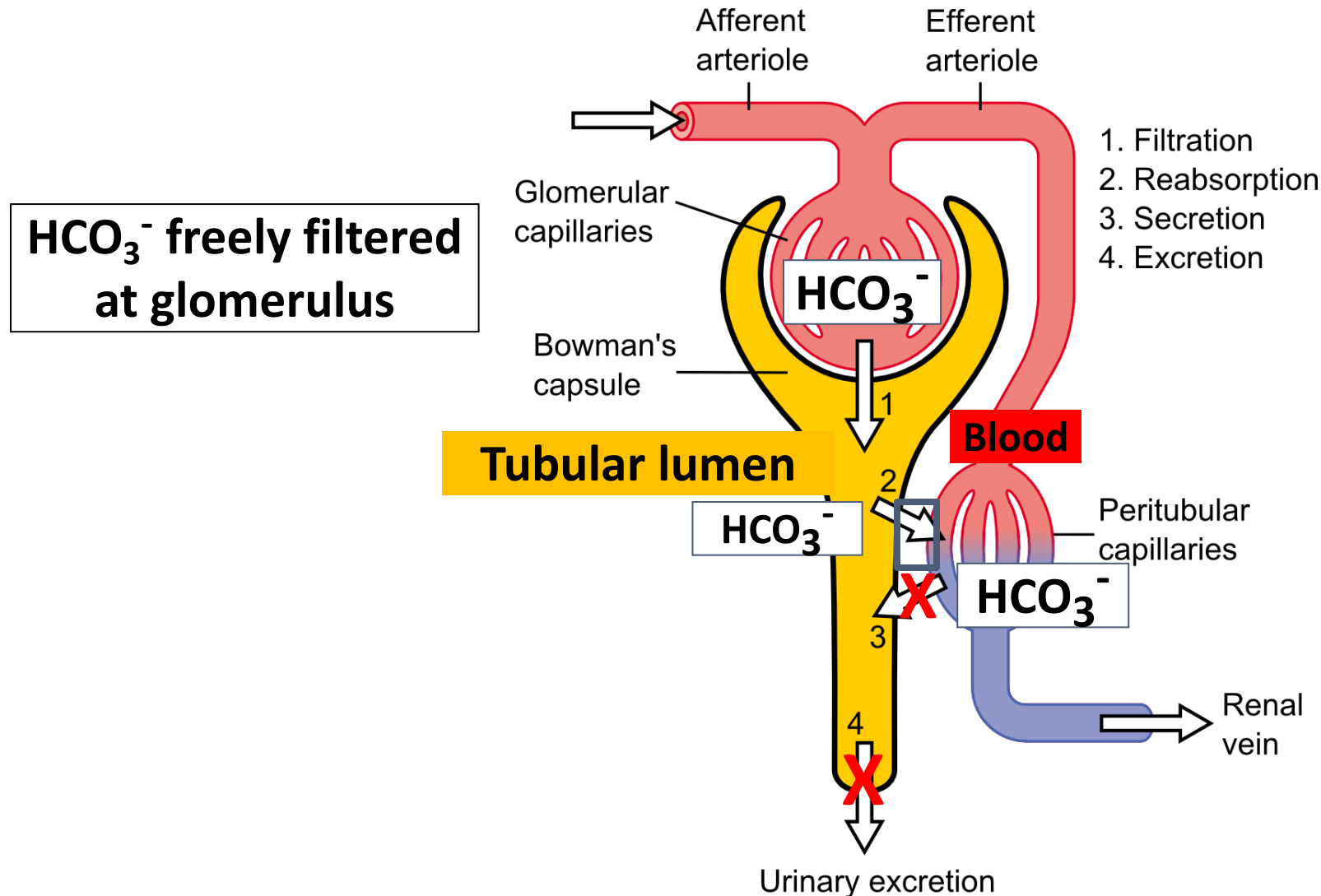
Recommended Reading

- Medical Physiology: Principles for Clinical Medicine, R. Rhoades & D. Bell; 6th ed.; Chapter 24, pp. 533-555; [Acid–Base Homeostasis | Medical Physiology: Principles for Clinical Medicine, 6e | Premium Basic Sciences | Health Library \(oclc.org\)](#)
- Costanzo Physiology; <https://www.clinicalkey.com/student/content/book/3-s2.0-B978032379333900016X>
- Ref: Mark's Essentials of Biochemistry: Chapter 4; [Water, Acids, Bases, and Buffers | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library \(oclc.org\)](#)
- Reference: Vander's Renal Physiology; Chapter 9: Acid base balance; [Regulation of Acid-Base Balance | Vander's Renal Physiology, 10th Edition | AccessMedicine | McGraw Hill Medical \(oclc.org\)](#)
- Additional problems: <http://www.scribd.com/doc/6442924/ABG-Interpretation>
Slide 17 onwards
- http://www.mhhe.com/biosci/ap/ap_casestudies/cases/ap_case14.html
- Practice questions folder for the week; Small group case

Role of Kidney in Acid-base balance

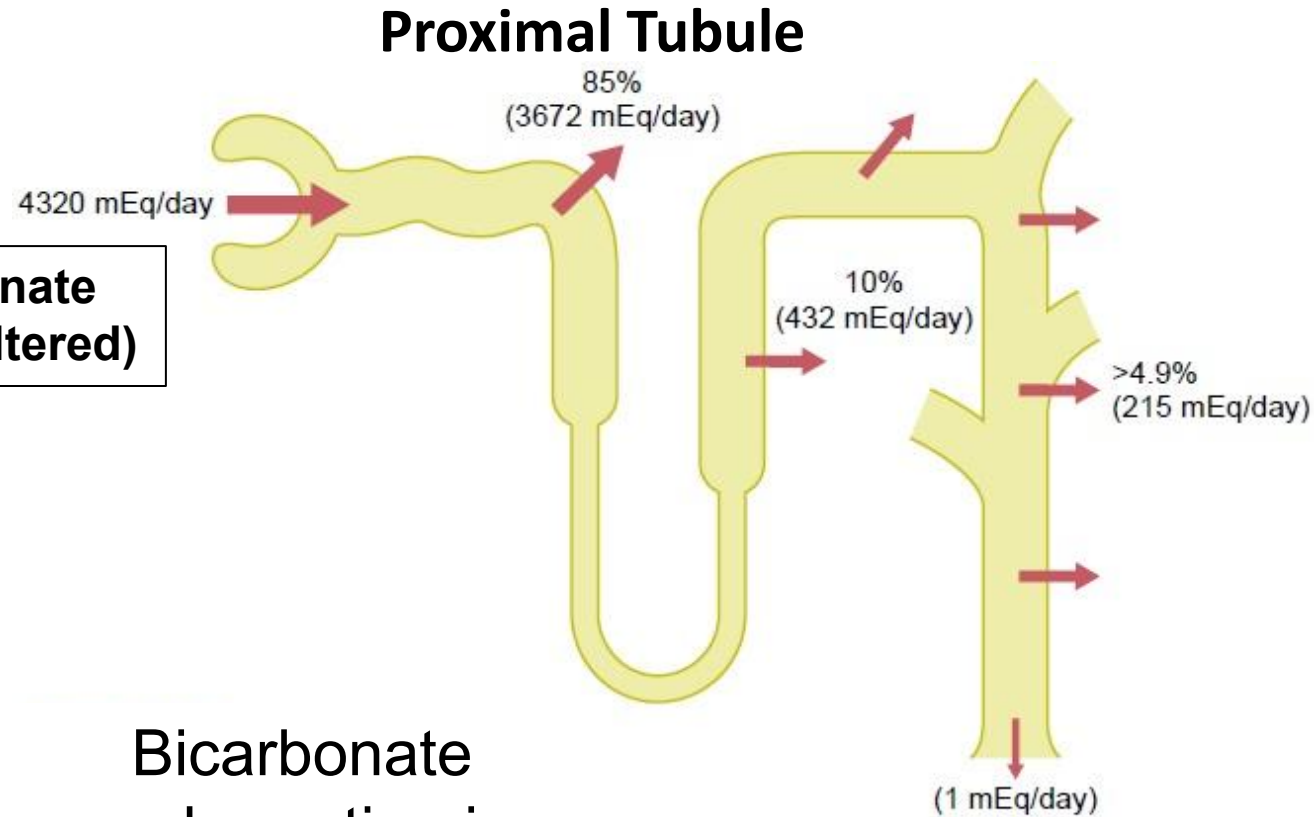
- Kidneys regulate **plasma HCO_3^-**
- Urine pH is 5.8-6 (acidic relative to plasma pH)
- Filtered HCO_3^- completely **reabsorbed**
 - HCO_3^- freely filtered at glomerulus
- Kidneys **secrete H^+**
 - Protons accepted by urinary buffers (phosphate and ammonia)
- HCO_3^- can also be '**newly**' formed in renal tubules
 - Replenish HCO_3^- lost by buffering nonvolatile acids (sulfate or lactic acid)
- Excrete sulfate and phosphate (non-volatile acids)

Overview of nephron and reabsorption of bicarbonate



SOM.MK.I.BPM1.3.CPR.1.INTG. 0010

Overview of a nephron and reabsorption of bicarbonate

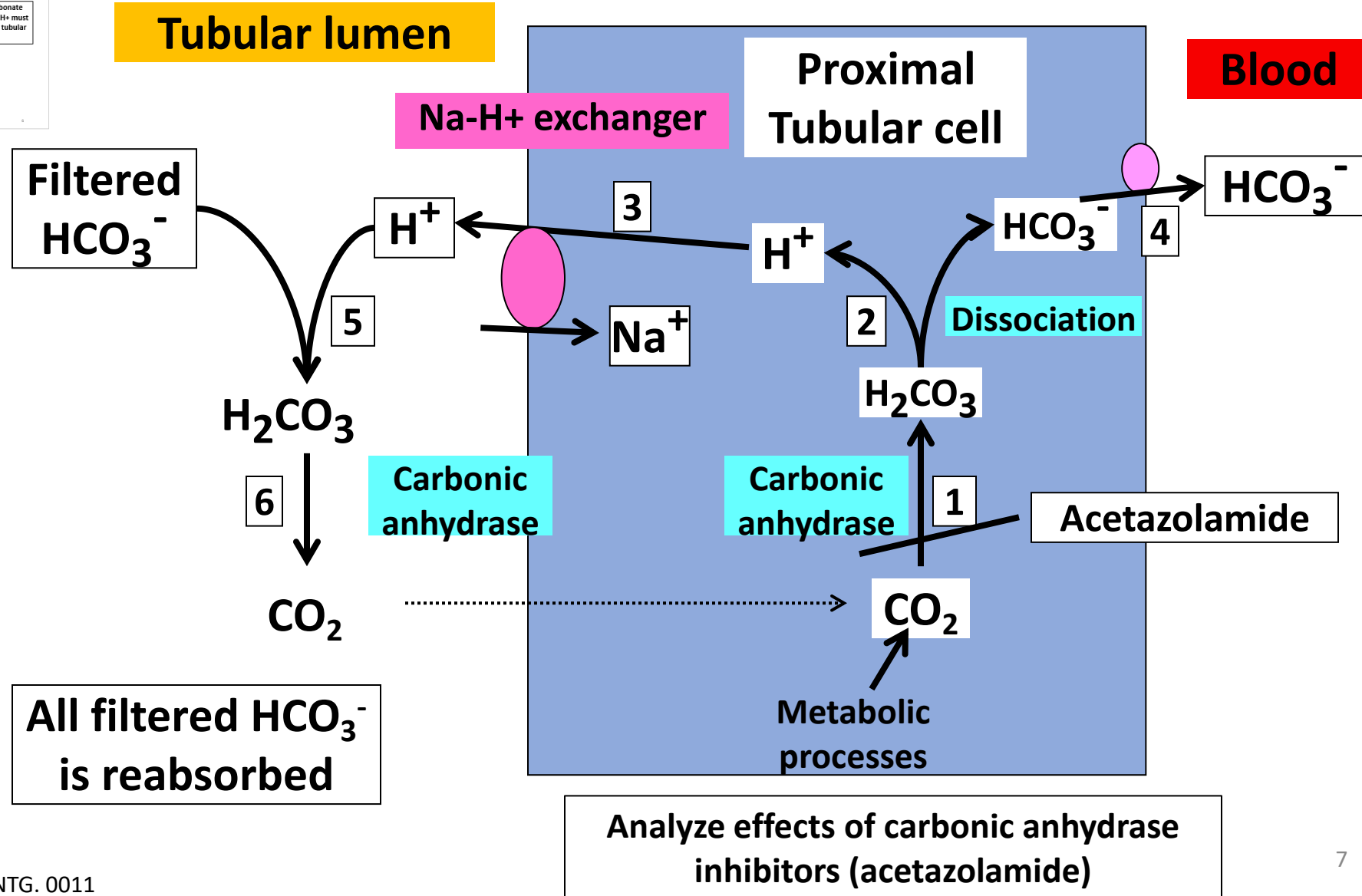
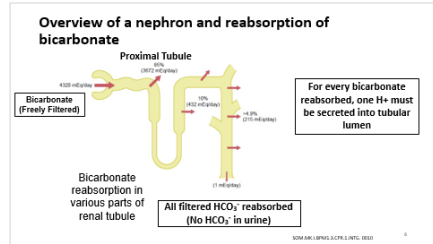


**Bicarbonate
(Freely Filtered)**

**For every bicarbonate
reabsorbed, one H⁺ must
be secreted into tubular
lumen**

**All filtered HCO₃⁻ reabsorbed
(No HCO₃⁻ in urine)**

Reabsorption of filtered bicarbonate

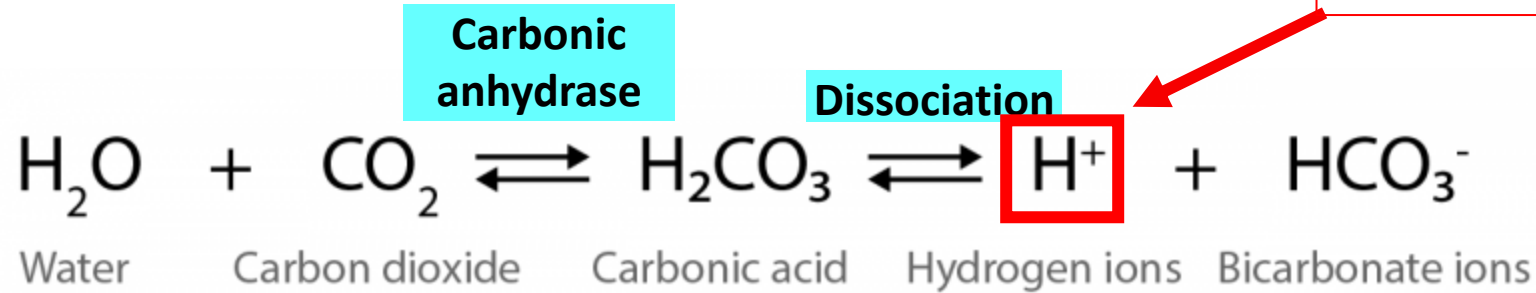


Reabsorption of filtered bicarbonate

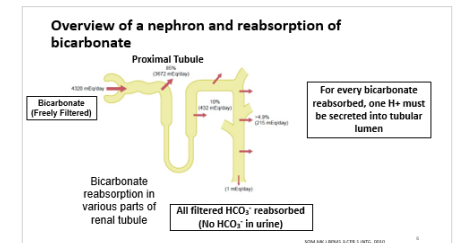
1. Formation of carbonic acid (H_2CO_3) by carbonic anhydrase in tubular cell
2. Dissociation of carbonic acid (weak acid) to bicarbonate and proton
3. Secretion of proton into tubular lumen (via $\text{Na}^+\text{-H}^+$ exchanger)
4. To maintain electrical neutrality, bicarbonate enters blood (For every proton secreted into the lumen, one bicarbonate enters blood)
5. Secreted proton associates with **filtered bicarbonate** in tubular lumen to form carbonic acid
6. Carbonic anhydrase in brush border of tubular cells converts Carbonic acid to CO_2 (diffuses into tubular cell to be reused)
7. 85% of filtered bicarbonate reabsorbed in **proximal tubule**; rest in distal tubule

When secreted proton buffered by bicarbonate buffer in tubular lumen = **Reabsorption** of filtered bicarbonate

Formation of 'NEW' bicarbonate

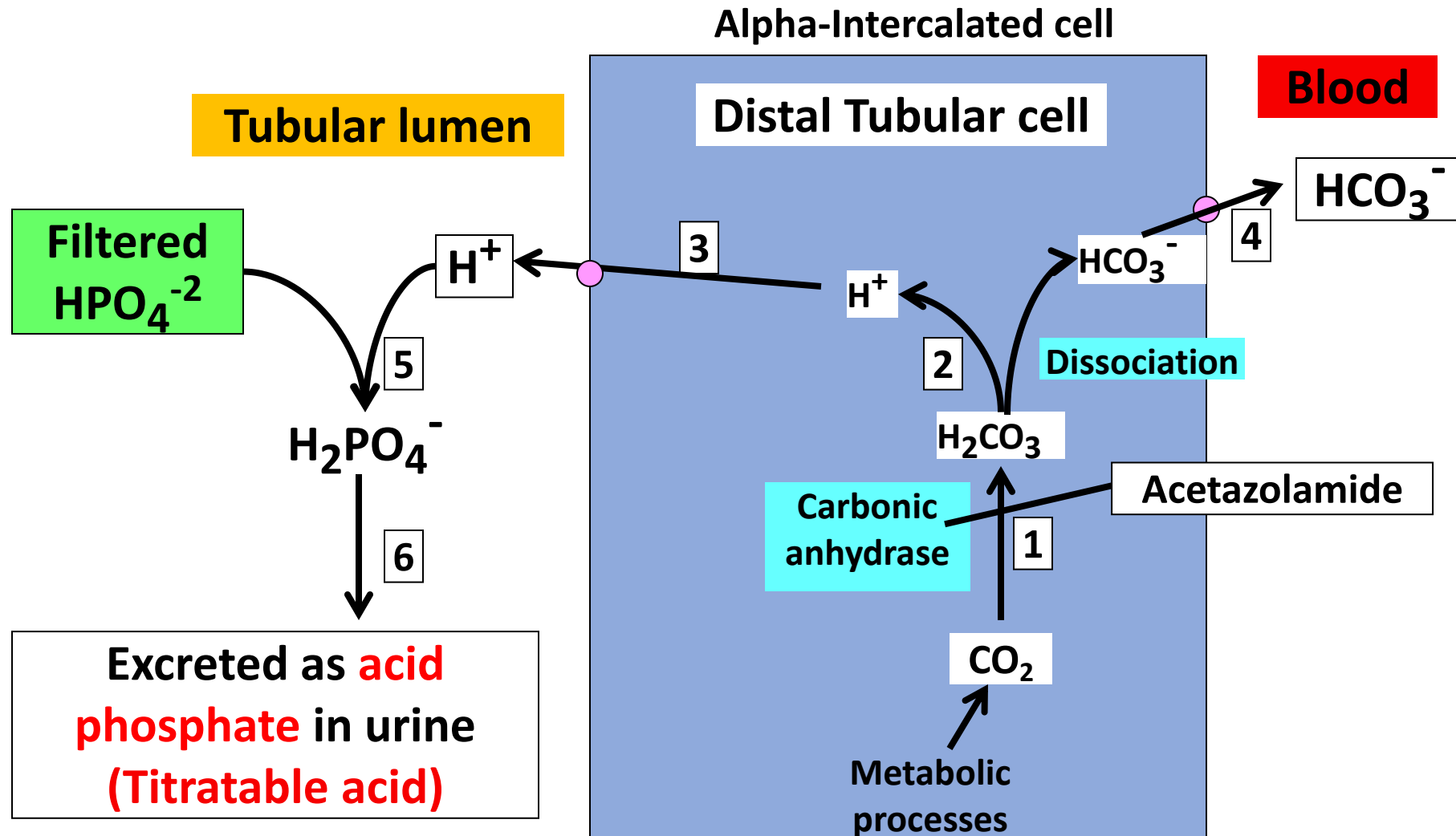


- When H^+ ions (non-volatile/ fixed acid) added, HCO_3^- used to buffer H^+
 - HCO_3^- levels fall
- 50-100 mEq non-volatile acid/day
- If kidney does NOT make new bicarbonate, serum HCO_3^- levels decrease
- Two mechanisms form NEW bicarbonate (Non-bicarbonate buffers): Active in **collecting duct (Distal nephron) (Alpha-intercalated cells)**
 - Phosphate buffer system
 - Ammonia buffer system



H⁺ Secretion: Role of phosphate buffer

Formation of 'NEW' bicarbonate



Secretion of H^+ : Role of phosphate buffer

Formation of 'NEW' bicarbonate

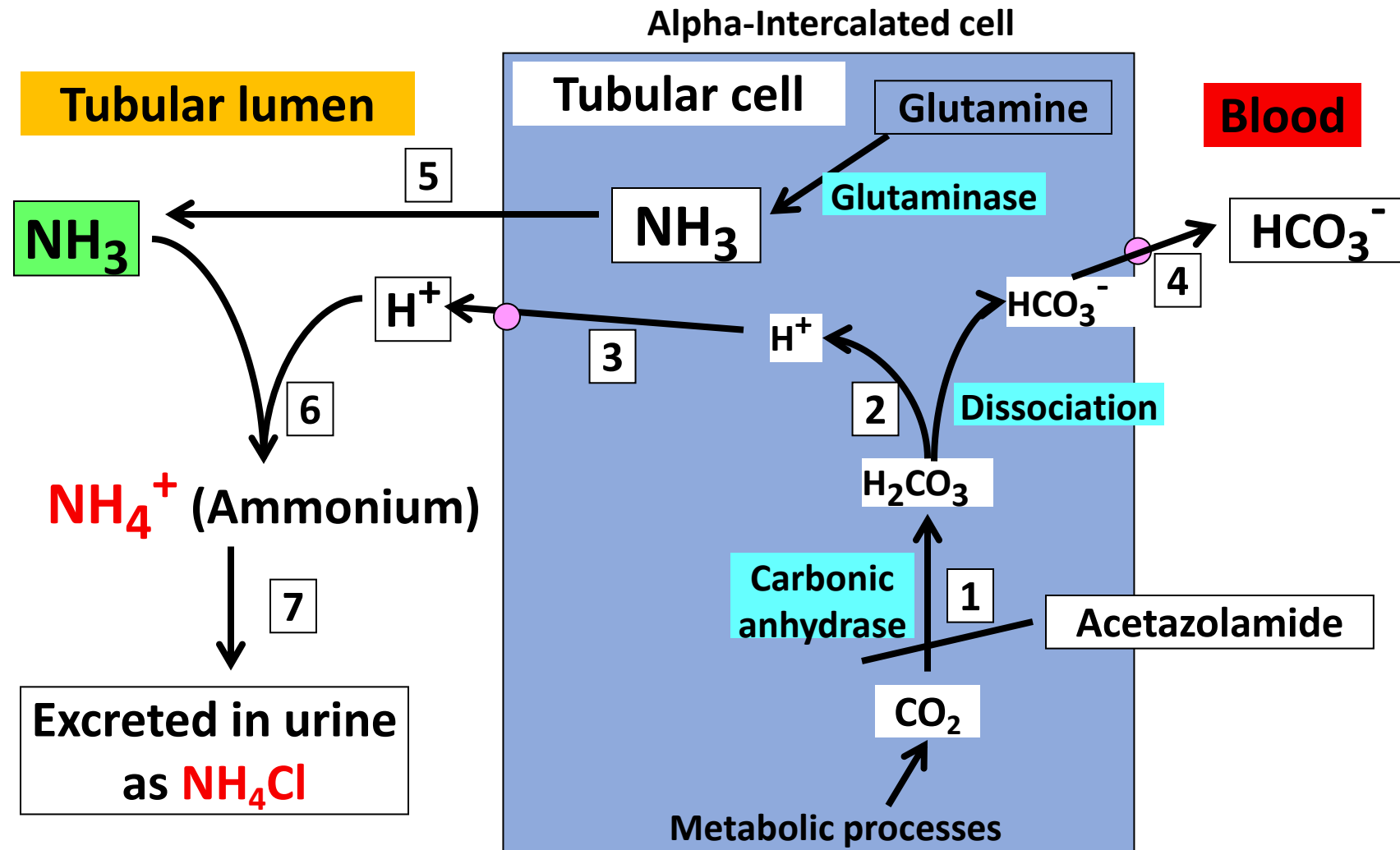
1. Formation of carbonic acid (H_2CO_3) by carbonic anhydrase in tubular cell
2. Dissociation of carbonic acid (weak acid) to bicarbonate and proton
3. Secretion of H^+ into tubular lumen
4. To maintain electrical neutrality, a bicarbonate enters blood (For every proton secreted into lumen, a bicarbonate enters the blood)
5. Proton associates with filtered phosphate to form acid phosphate (alpha-intercalated cells in distal tubule)
6. Acid phosphate excreted in urine and contributes to **titratable acidity** in urine

When secreted proton buffered by a non-bicarbonate buffer (phosphate/ ammonia) in tubular lumen = Formation of **NEW** bicarbonate

Secretion of H^+ : Role of ammonia

Formation of 'NEW' bicarbonate

Tubular capacity to form ammonium is **unlimited** and highly stimulated in **prolonged acidosis**



Secretion of H^+ : Role of ammonia

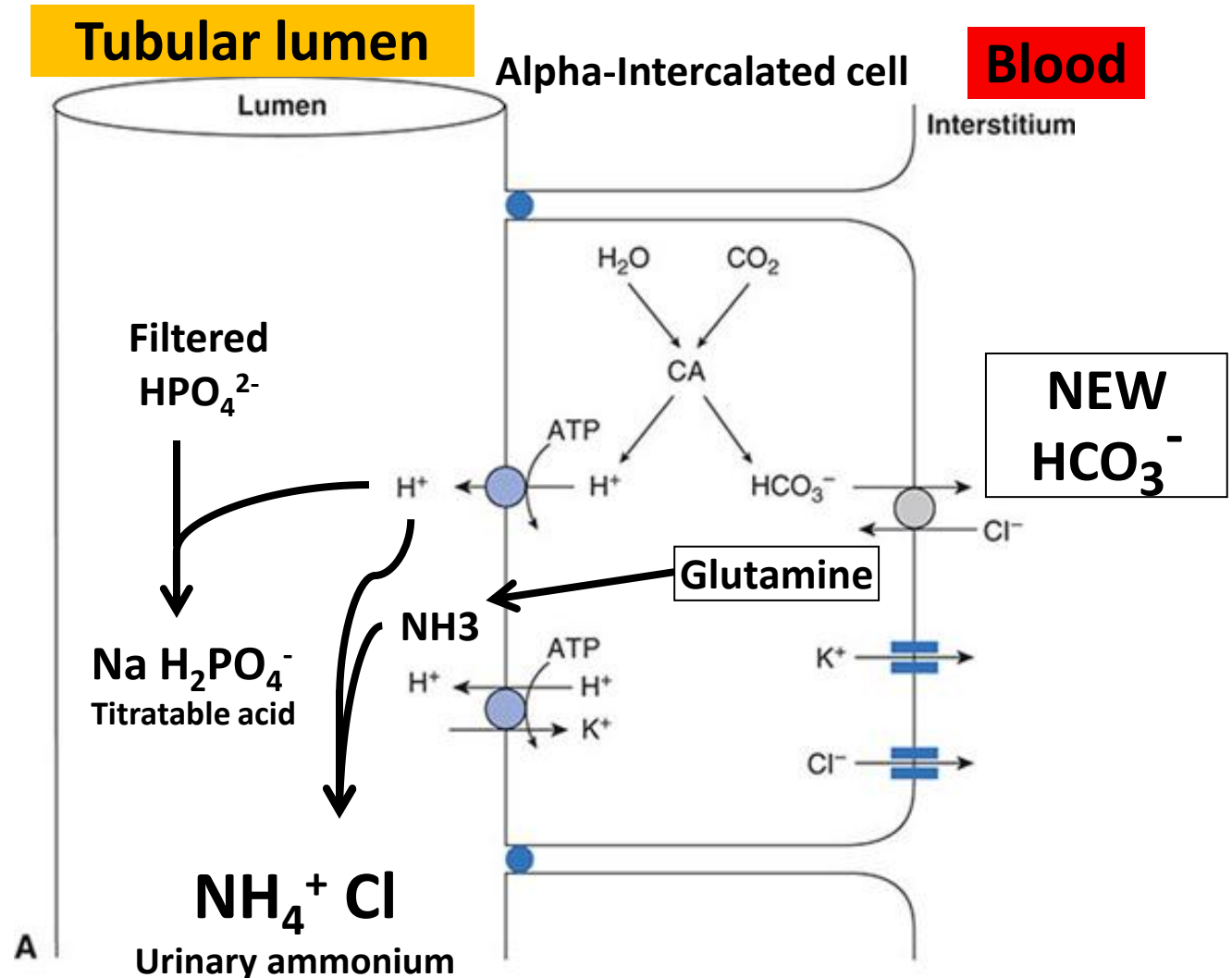
Formation of '**NEW**' bicarbonate

1. Formation of carbonic acid (H_2CO_3) by carbonic anhydrase in tubular cell
2. Dissociation of carbonic acid (weak acid) to bicarbonate and proton
3. Secretion of H^+ into tubular lumen
4. To maintain electrical neutrality, a bicarbonate enters blood (For every proton secreted into lumen, a bicarbonate enters blood)
5. Hydrolysis of glutamine by **glutaminase** forms ammonia in renal tubule
 - NH_3 is lipid soluble and diffuses into tubular lumen
 - **Hypokalemia** stimulates glutaminase
6. NH_3 binds to secreted proton in tubular lumen to form Ammonium ion (NH_4^+) ($pK=9.2$) → Trapped in lumen (**Distal tubule and collecting duct**)
7. NH_4^+ excreted as **ammonium chloride** in urine
8. Tubular capacity to excrete ammonium is **unlimited** and highly stimulated in prolonged acidosis

When secreted proton buffered by a non-bicarbonate buffer (phosphate/ ammonia) in tubular lumen = Formation of **NEW** bicarbonate

H⁺ secretion in Collecting duct

- H⁺ ATPase
- K⁺-H⁺ ATPase
- Secreted protons buffered by:
 - HPO₄⁻ forms Acid Phosphate (NaH₂PO₄⁻)
 - NH₃ forms NH₄⁺ Cl (Urinary ammonium chloride/sulfate)

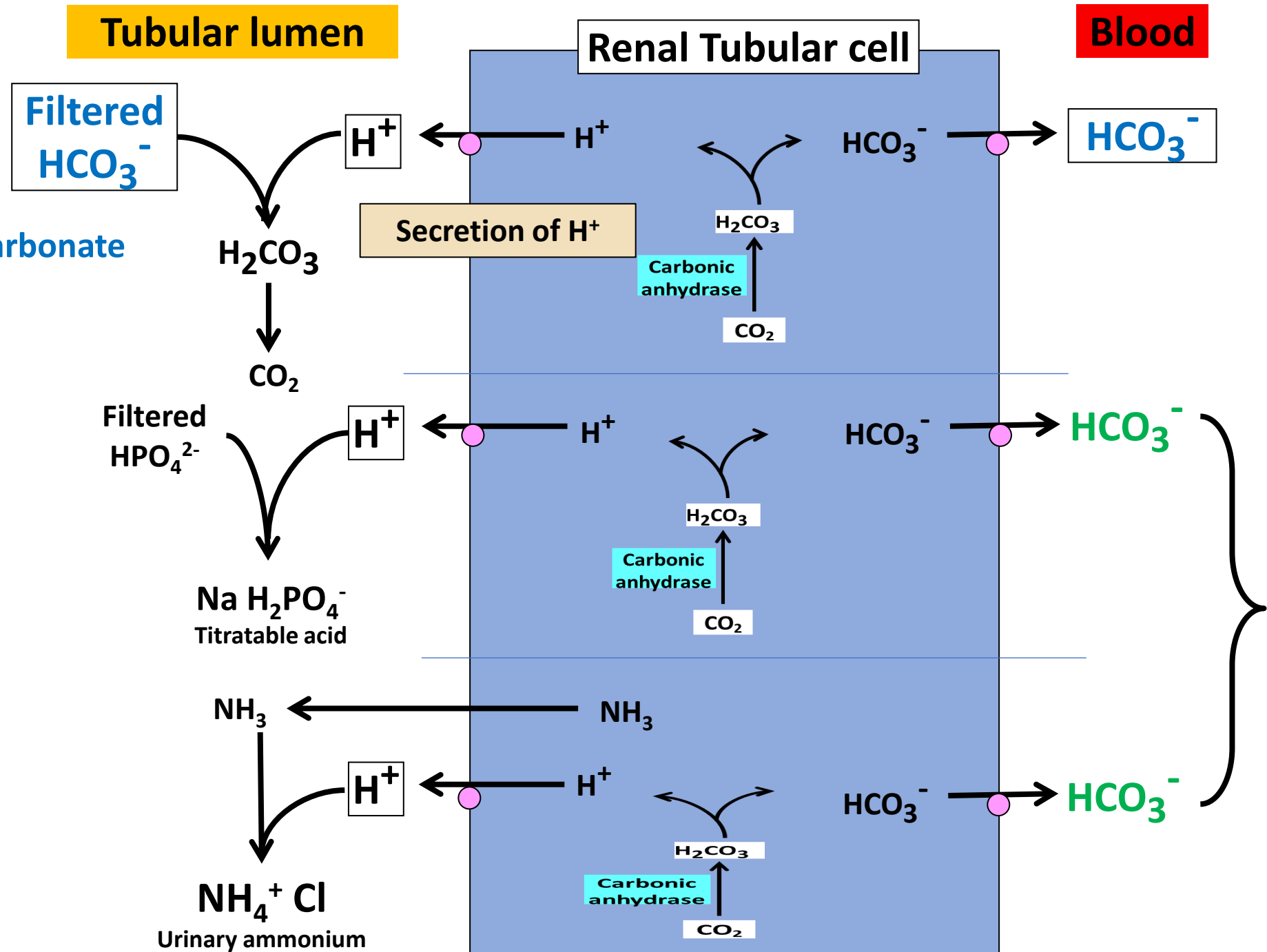


Reabsorption of filtered bicarbonate

Summary

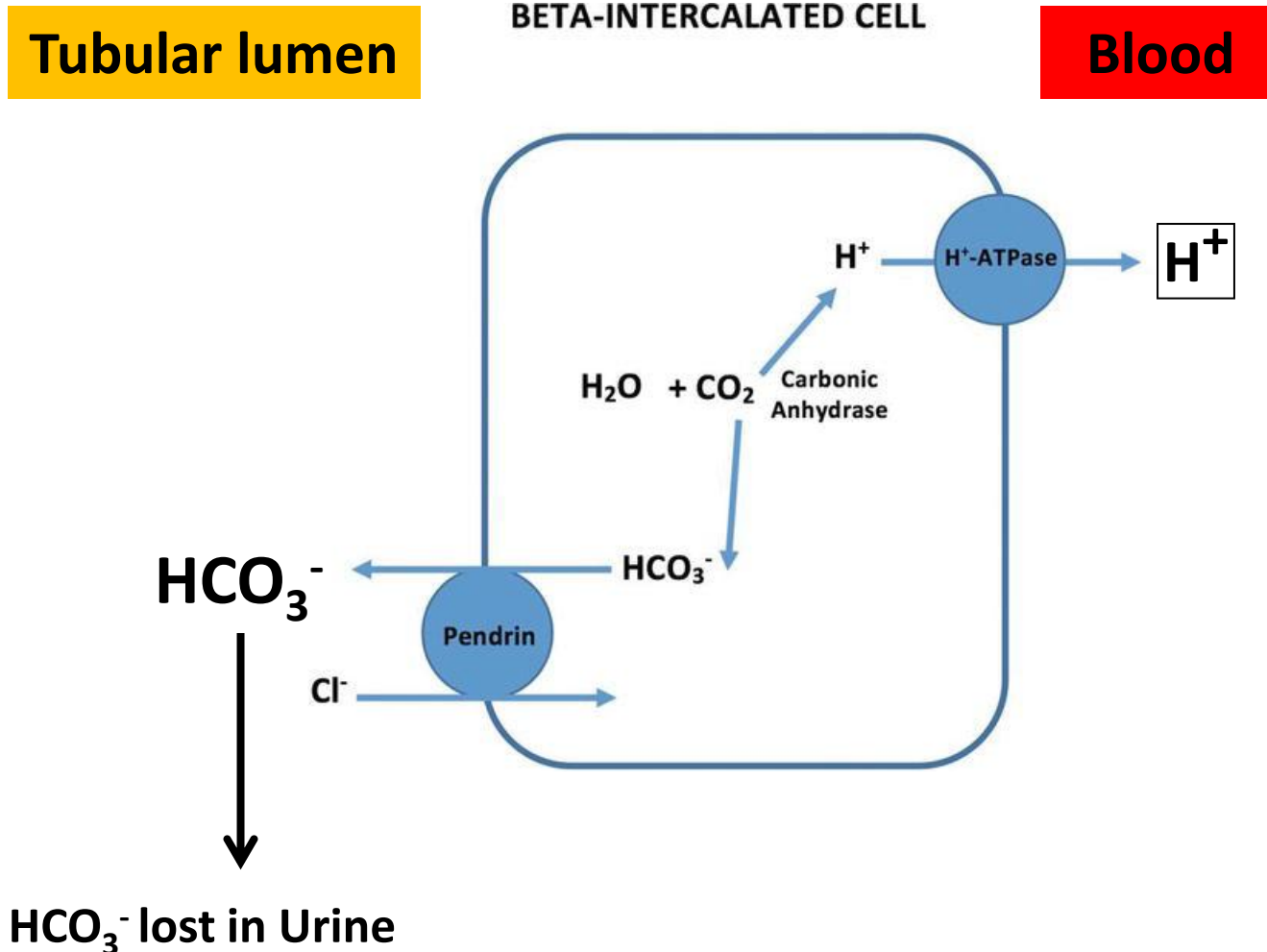
Formation of
'NEW' bicarbonate

Effect of:
Chronic acidosis
Acetazolamide



Bicarbonate secretion in urine

Beta-Intercalated cells (Collecting duct)



- Secrete HCO_3^- in urine
- Reabsorb H^+
- Makes urine alkaline
- Active in **chronic alkalosis**

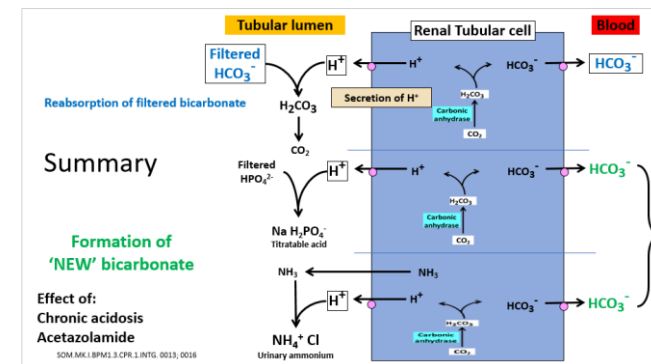
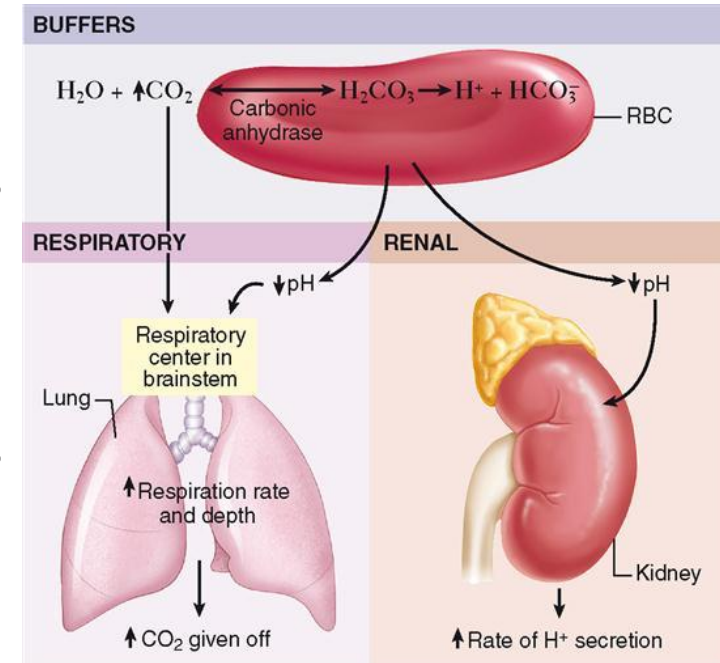
Quantification of urinary acid loss

- H^+ in urine = Acid phosphate (Titratable acidity) + Ammonium – Urinary Bicarbonate
- Chronic acidosis → Increased H^+ secretion in tubules and increased **urinary acid phosphate** and **Ammonium excretion**
- Chronic alkalosis → Decreased H^+ secretion in tubules and decreased urinary acid phosphate and ammonium. **Tubules secrete bicarbonate** which is lost in urine

Condition	Total Production of Fixed H^+ (mEq/day)	Excretion of H^+ as Titratable Acid (mEq/day)	Excretion of H^+ as NH_4^+ (mEq/day)
Normal	50	20	30
Diabetic ketoacidosis	500	100	400
Chronic kidney disease	50	10	10

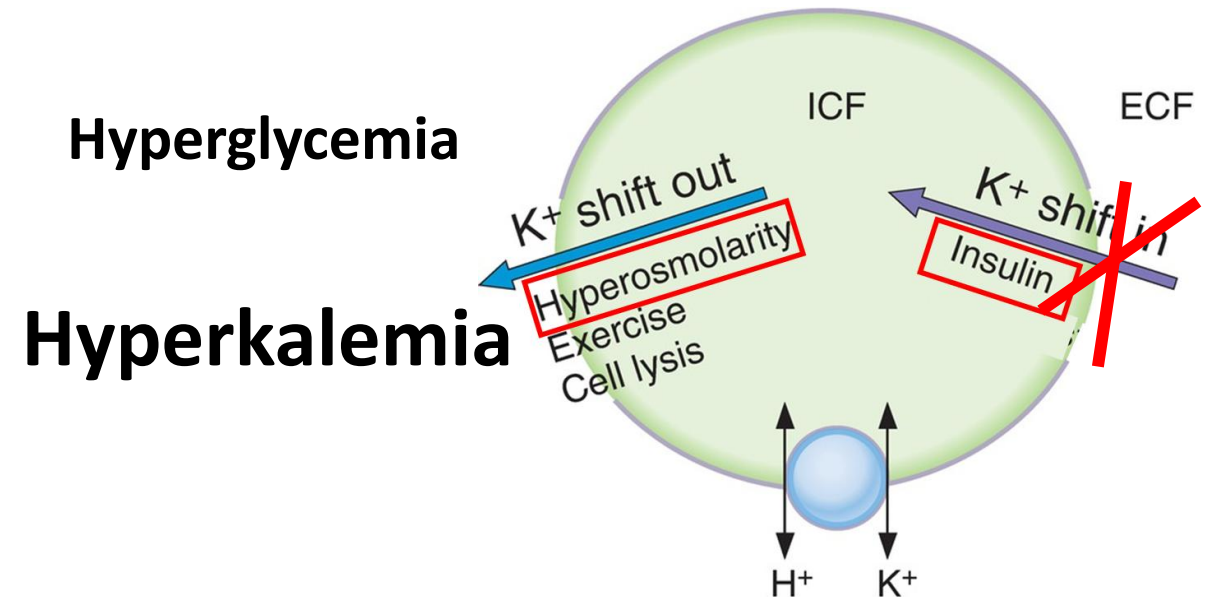
Factors that increase H^+ secretion in renal tubules

- Increased $PaCO_2$ (Respiratory acidosis) → Increased carbonic anhydrase activity and increased urinary H^+ secretion (excretion)
- Decreased pH (Metabolic acidosis) → Increased carbonic anhydrase activity and increased urinary H^+ secretion (excretion)
- Hypokalemia
- Increased aldosterone
- Decreased ECF volume
- Thiazide and loop diuretics (increase sodium load in renal tubules)



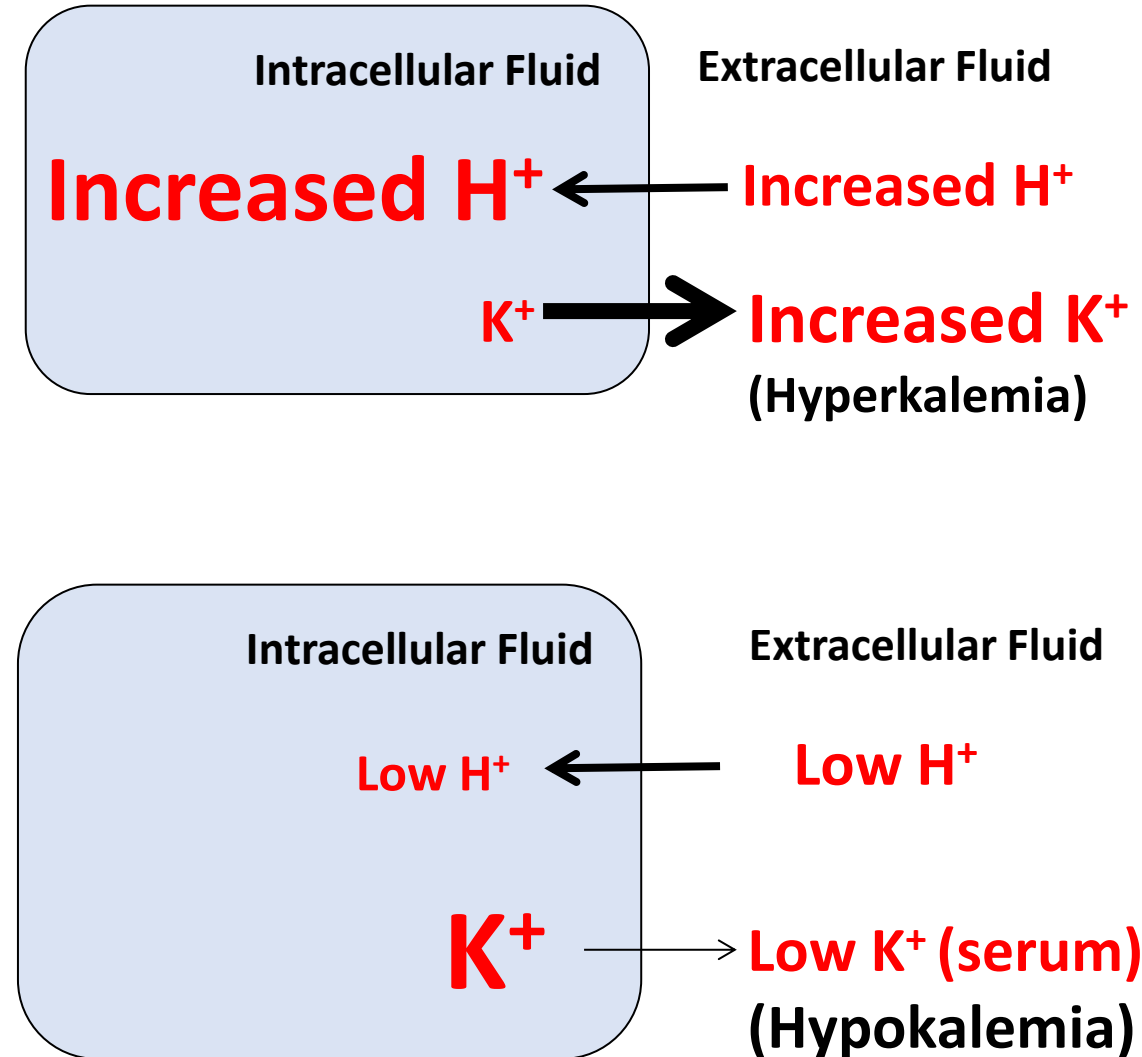
H⁺ and K⁺ interrelations

- K⁺ (major intracellular cation)
- **Diabetic ketoacidosis:**
 - Plasma glucose: 400mg/dL; Serum K⁺: 5.6 mEq/L (High)
 - **Insulin deficiency**
 - Insulin activates Na⁺-K⁺ ATPase
 - **Insulin deficiency: Na⁺-K⁺ pump less active**
 - **Hyperglycemia** → **Hyperosmolarity** → K⁺ shift from ICF to ECF (with water)
 - → Increased serum K⁺ (Hyperkalemia)



Metabolic acid-base disorders → Serum Potassium changes

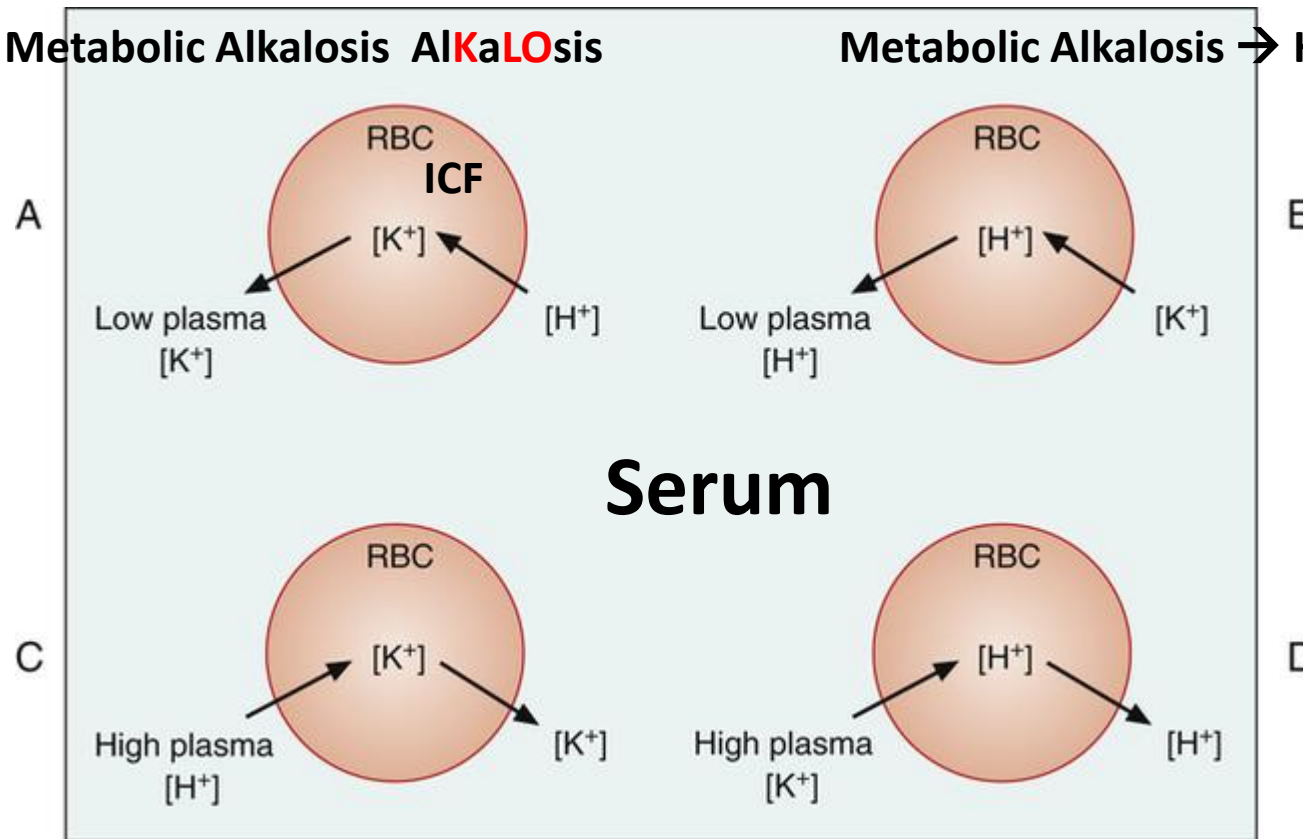
- Changes in pH affect serum (ECF) K^+ levels
 - Metabolic Acidosis – Hyperkalemia
 - Metabolic Alkalosis – Hypokalemia
- In metabolic acidosis, excess H^+ from ECF enters ICF
- ICF K^+ moves to ECF (Hyperkalemia)
- **Metabolic Acidosis** accompanied by hyperkalemia
- **Metabolic Alkalosis:** K^+ retained in ICF (hypokalemia): **AlKaLOsis**



Metabolic acid-base disorders $\Leftrightarrow$ Serum Potassium changes

Hypokalemia $\rightarrow$ Metabolic Alkalosis **AlKaLOsis**

Metabolic Alkalosis $\rightarrow$ Hypokalemia **AlKaLOsis**

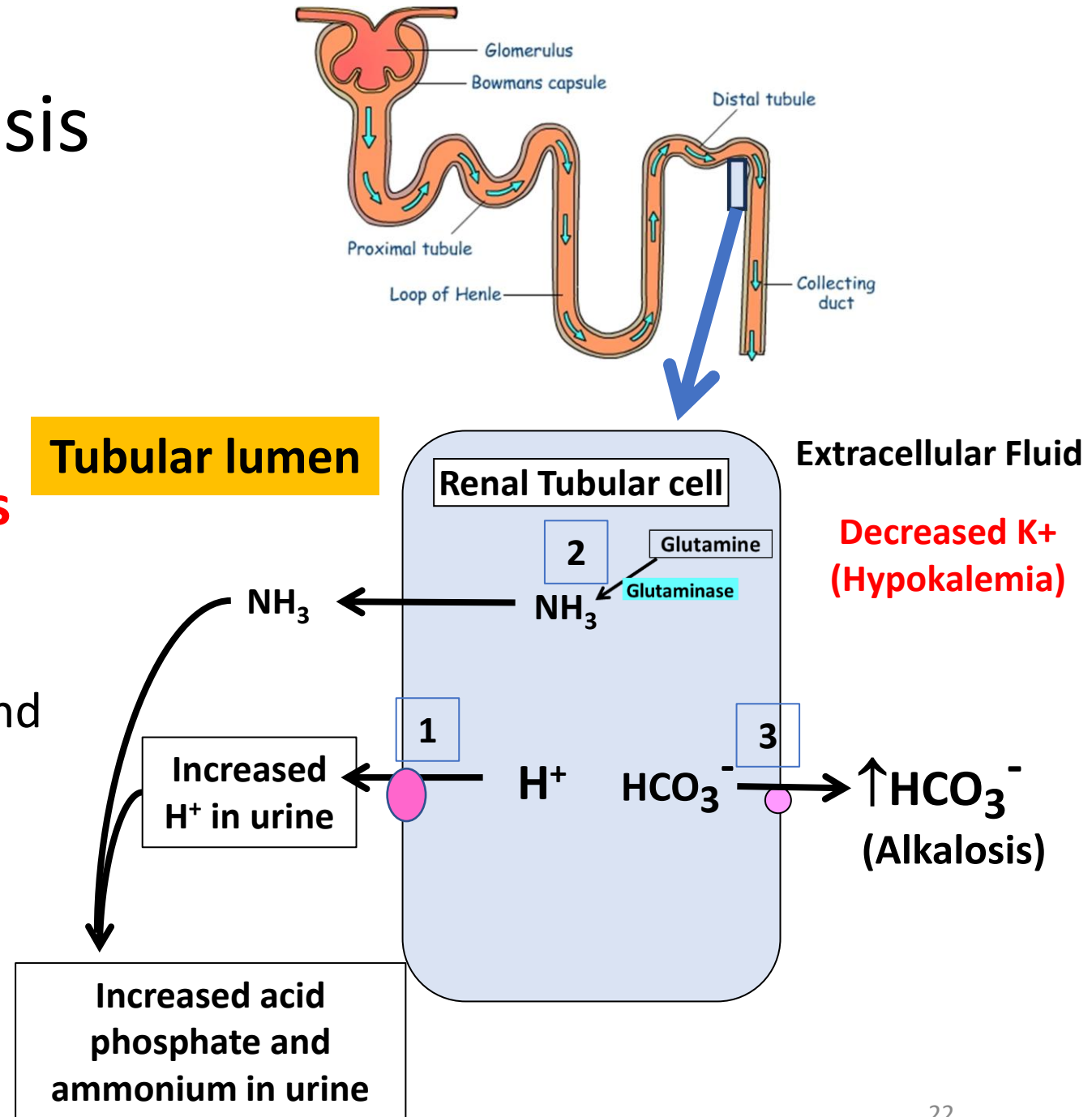


Metabolic Acidosis $\rightarrow$ Hyperkalemia

Hyperkalemia $\rightarrow$ Metabolic Acidosis

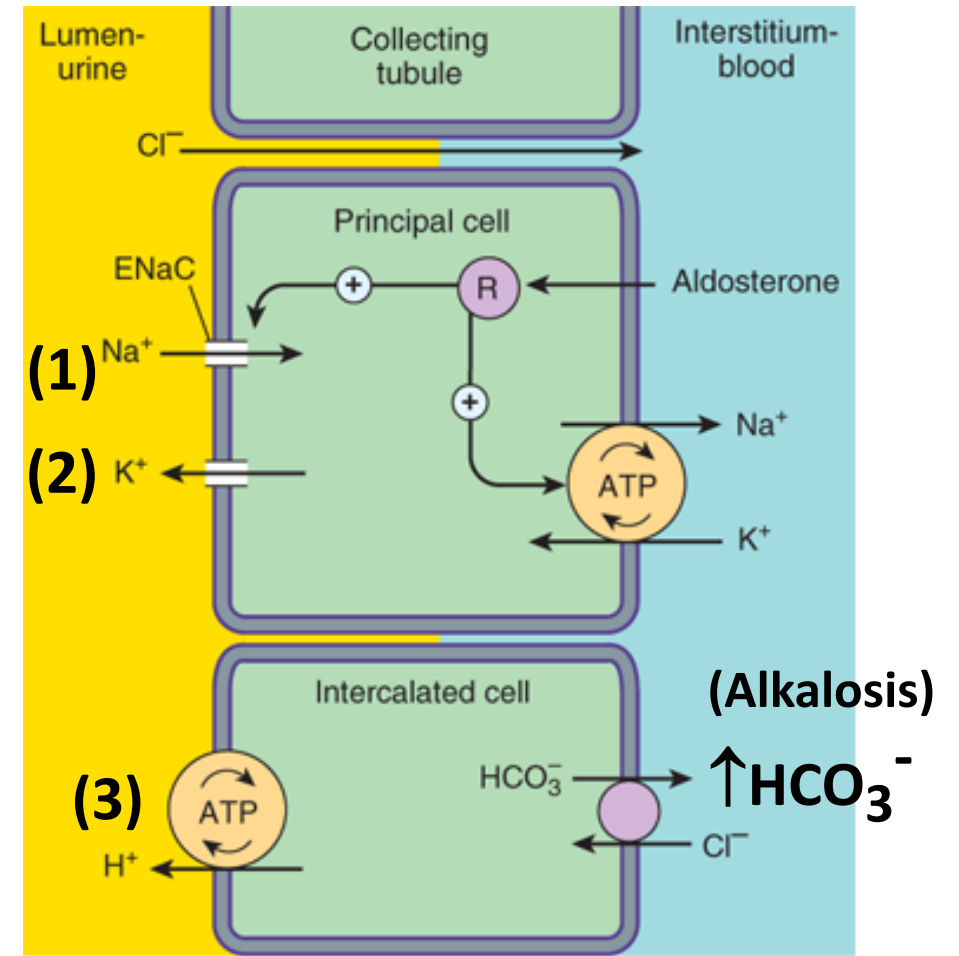
Hypokalemia → Alkalosis (Renal tubules)

- Changes in serum K^+ affects pH
 - Hypokalemia → Alkalosis
- Hypokalemia: Effect on **renal tubules**
 1. Stimulates H^+ secretion into lumen
 2. Stimulates ammonia formation
 3. Stimulates formation of new HCO_3^- and increases serum HCO_3^- (Alkalosis)
- Hypokalemia worsens alkalosis
- **TIP:** Treat hypokalemia to resolve alkalosis



Primary Hyperaldosteronism (Conn syndrome)

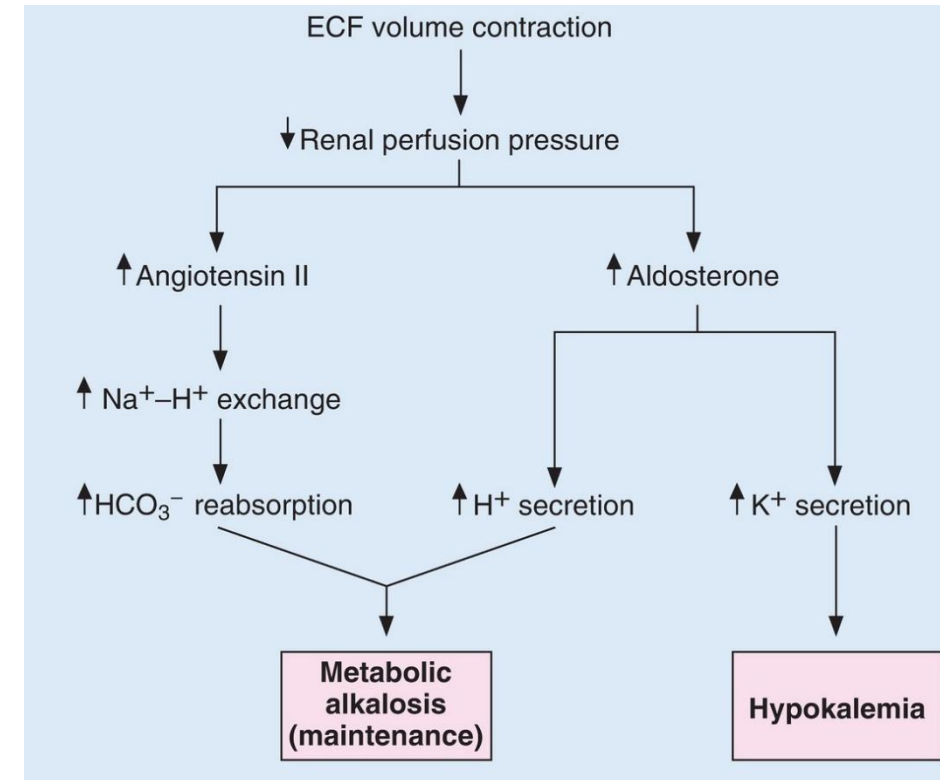
- Aldosterone stimulates reabsorption of Na^+ (Activates ENaC $\rightarrow$ Increases serum Na^+ levels and blood volume $\rightarrow$ **Hypertension**) (1)
- Stimulates K^+ excretion in urine (2) (**Hypokalemia**)
- Stimulates renal tubular H^+ ATPase (3)
 - Increased tubular H^+ secretion
 - Increases HCO_3^- gain by blood $\rightarrow$ Alkalosis



Source: A.J. Trevor, B.G. Katzung, M. Kruidering-Hall: Katzung & Trevor
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Decreased ECF volume (Volume contraction)

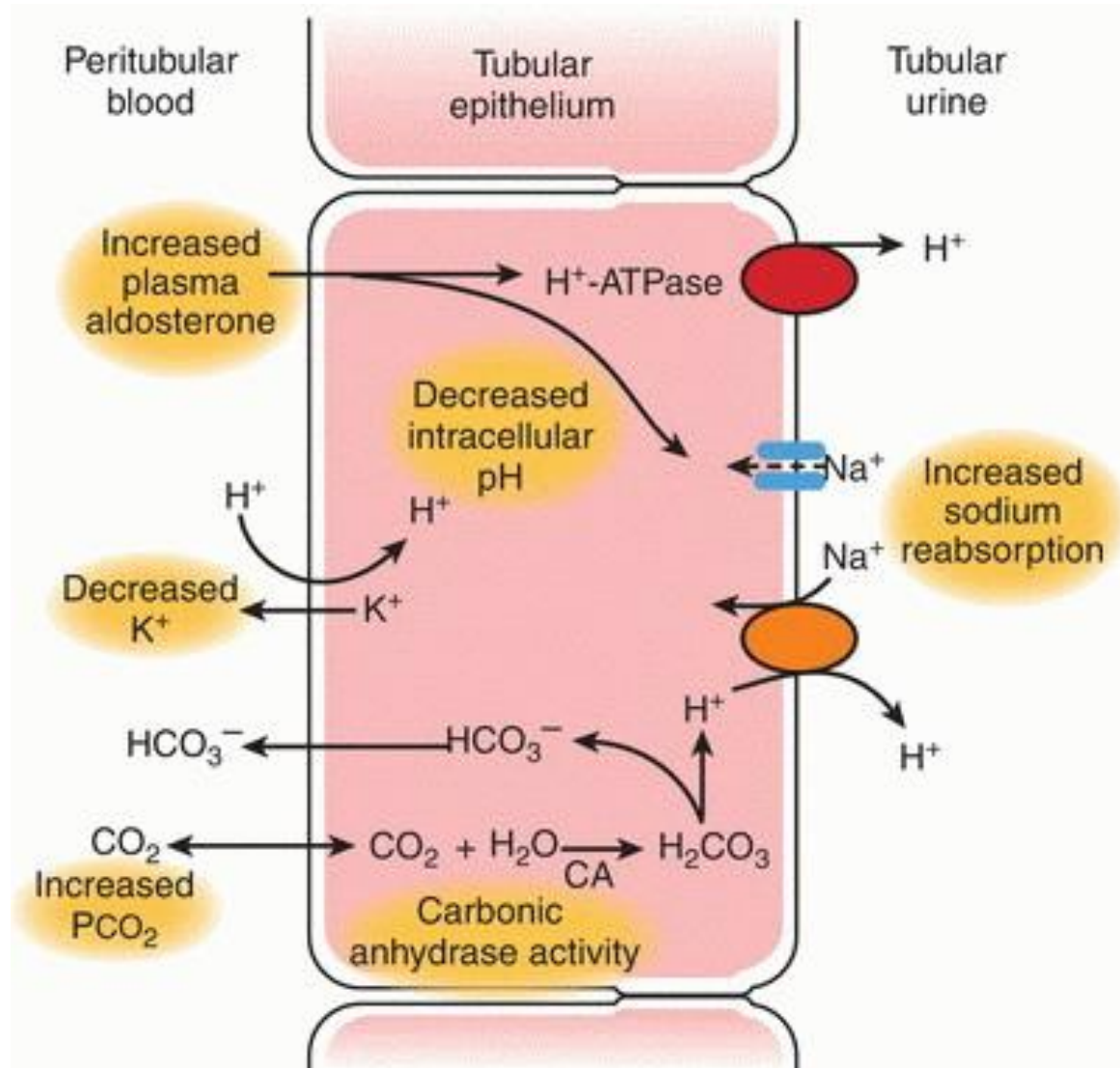
- Low blood volume → Stimulates **RAAS**
- Angiotensin-II increases $\text{Na}^+\text{-H}^+$ antiporter to increase Na reabsorption (H^+ secreted → increased reabsorption of HCO_3^-)
- **Aldosterone** increases sodium reabsorption + Cl^- and H_2O reabsorption; **Stimulates H^+ secretion**
- **Stimulates H^+ secretion** (ENaC reabsorbs Na^+ and lumen becomes negative → H^+ secretion by alpha-intercalated cell)
- Volume contraction and **metabolic alkalosis** → Increases Na^+ reabsorption (to increase blood volume) and H^+ secretion in urine (paradoxic aciduria!!)
- **TIP:** Restore blood volume before treating alkalosis



Diuretics (Loop or thiazide)

- **Loop Diuretics** – Decrease Na^+ reabsorption in loop of Henle
 - Result = $\downarrow \text{Na}^+$ reabsorption $\rightarrow \uparrow \text{Na}^+$ delivery to the distal tubule
- **Thiazides** – Decrease Na^+ reabsorption in distal convoluted tubule
 - Result = Decreased NaCl reabsorption
- Both cause **ECF volume contraction** $\rightarrow$ Secretion of **aldosterone** and Angiotensin-II
- Decreased Na^+ reabsorption in tubules $\rightarrow \uparrow \text{Na}^+$ delivery to distal tubule $\rightarrow \uparrow \text{K}^+$ secretion and $\uparrow \text{H}^+$ secretion in urine $\rightarrow$ **Metabolic alkalosis and hypokalemia**

Factors affecting tubular H^+ secretion



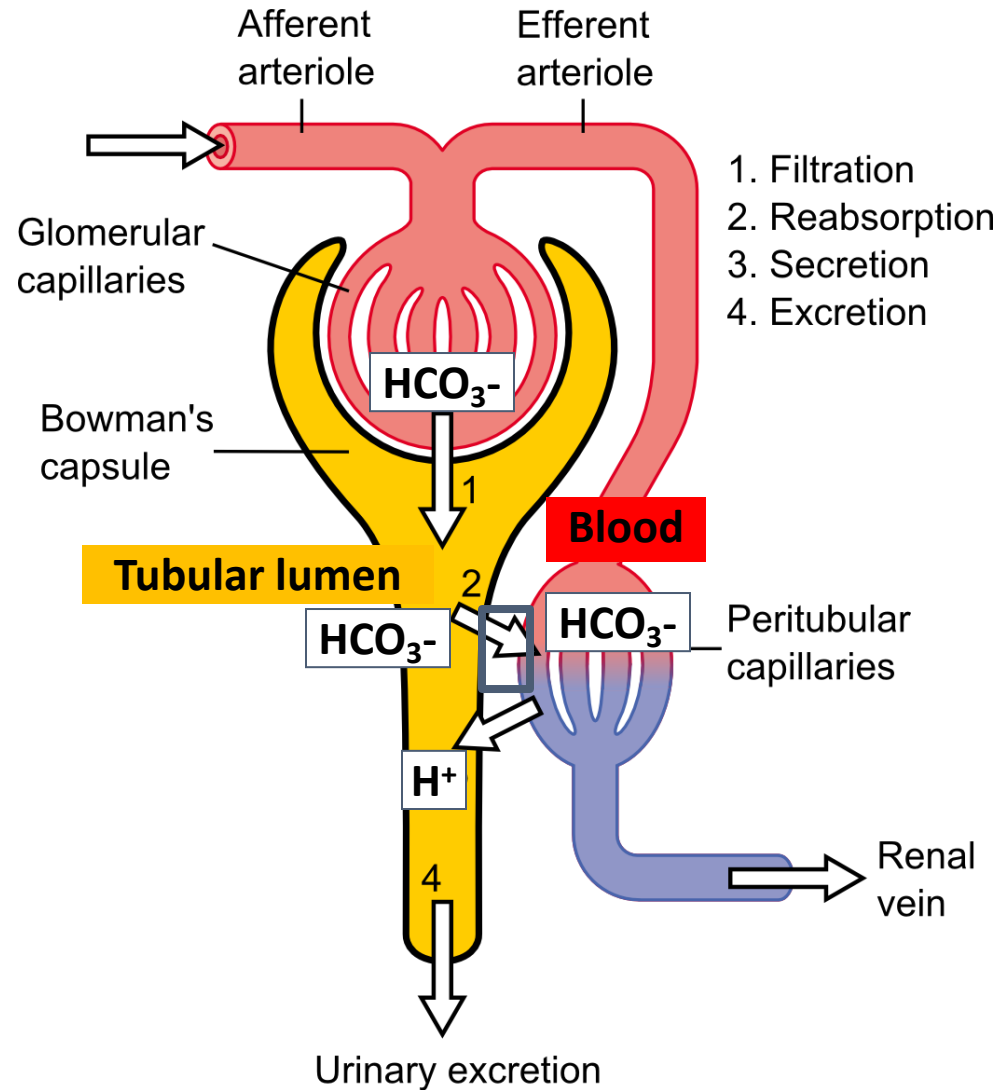
- **Increased tubular H^+ secretion**

- Decreased intracellular pH (acidosis)
- Increased $PaCO_2$ (respiratory acidosis)
- Increased plasma aldosterone
- Hypokalemia
- Increased Na^+ reabsorption (volume contraction)
- Thiazide and loop diuretics

- **Decreased tubular H^+ secretion**

- Inhibition of carbonic anhydrase (acetazolamide)
- Low $PaCO_2$ (Chronic respiratory alkalosis)
- Metabolic alkalosis (with normal blood volume)

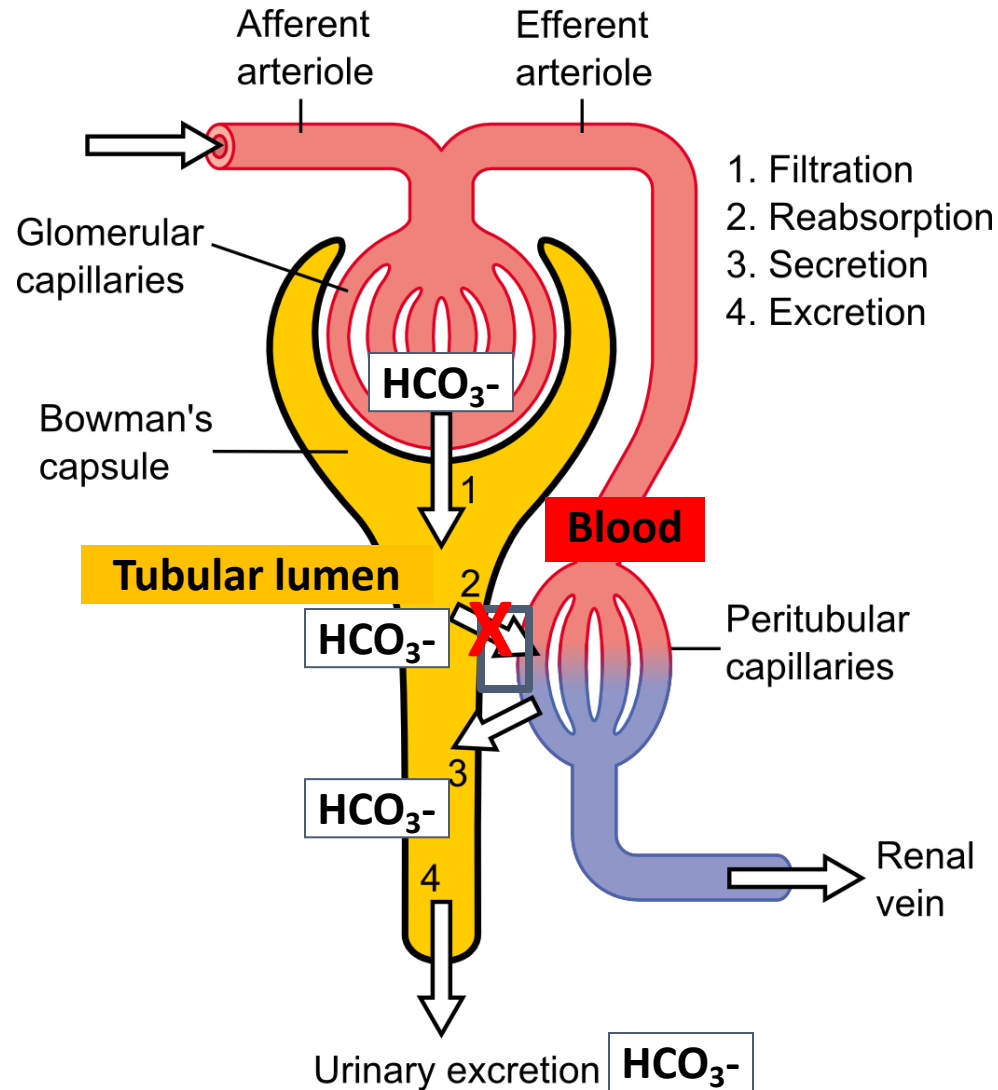
Renal compensation in respiratory/ metabolic acidosis



- Chronic acidosis (if kidney normal),
- Increased H^+ secretion by tubule
- Filtered bicarbonate is reabsorbed
- Urinary Acid phosphate (titratable acidity) is increased
- Ammonium in urine is increased
- Urine pH becomes acidic

Increased phosphate and ammonium excretion in urine

Renal compensation in respiratory/ metabolic alkalosis



Increased bicarbonate loss in urine

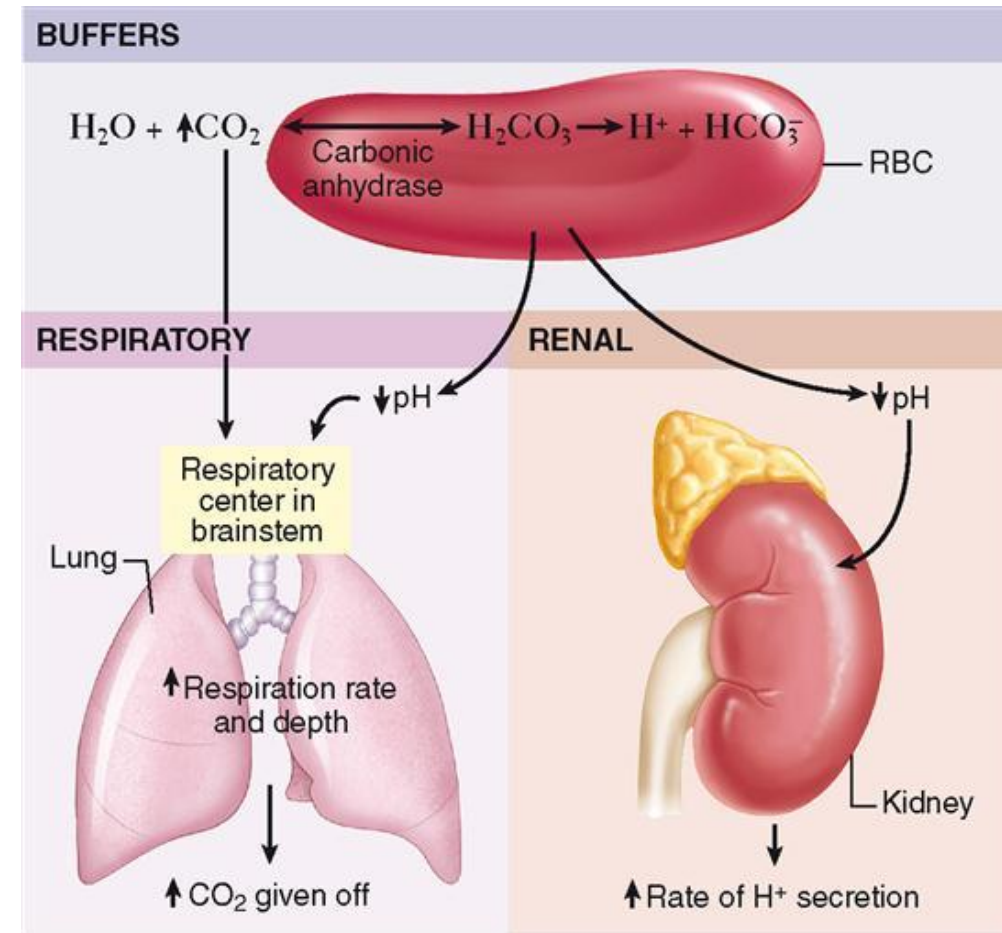
- Alkalosis (if kidney normal and fluid volume normal)
- Decreased H^+ secretion by tubule
- Filtered bicarbonate NOT reabsorbed
- Urinary bicarbonate loss increases
- Acid phosphate and ammonium in urine decreases
- Bicarbonate also secreted by tubular cells
- Urine pH becomes alkaline

Summary: Role of buffers

- Normal levels: PaCO_2 : 40 mmHg; HCO_3^- : 24 mEq/L
- Ratio $\text{HCO}_3^- : \alpha \text{PaCO}_2 = 20:1$
- Normal $\text{H}^+ = 40 \text{ nmol/L}$; pH: 7.4 (inversely related)
- **Buffers:** First line of defense against pH changes
 - Immediate acceptors of protons (H^+) generated during metabolism
 - Acid base disturbances measure pH, PaCO_2 and HCO_3^-

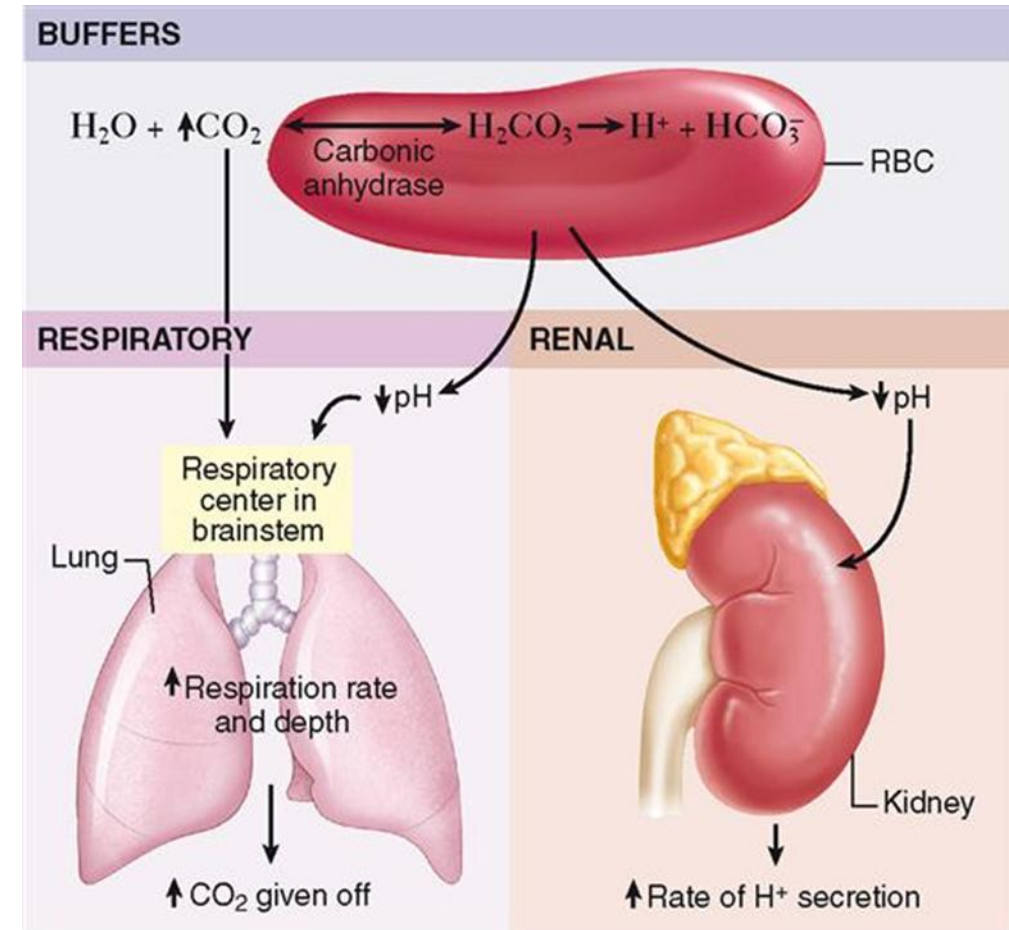
Summary: Role of Respiratory system (Regulates PCO_2)

- Respiratory system disorders results in respiratory acid-base disturbances
 - Decreased ventilation $\rightarrow$ Increased PCO_2
 - Increased ventilation $\rightarrow$ Decreased PCO_2
- Immediate response to pH changes (Immediate compensation)
- When pH falls (metabolic acidosis) $\rightarrow$ Increased respiratory rate and increased washout of CO_2 $\rightarrow$ Fall in PaCO_2
- When pH increases (metabolic alkalosis) $\rightarrow$ Decrease respiratory rate and increased retention of CO_2 $\rightarrow$ Increased PaCO_2



Summary: Role of Kidney (Regulates HCO_3^-)

- Responds to pH changes more slowly, but response is prolonged (sustained)
 - When pH falls (acidosis) → Increased proton secretion in urine and urine becomes acidic. Increased phosphate and ammonium (NH_4^+) excretion in urine
 - When pH increases (alkalosis) → Increased HCO_3^- excretion and urine becomes alkaline

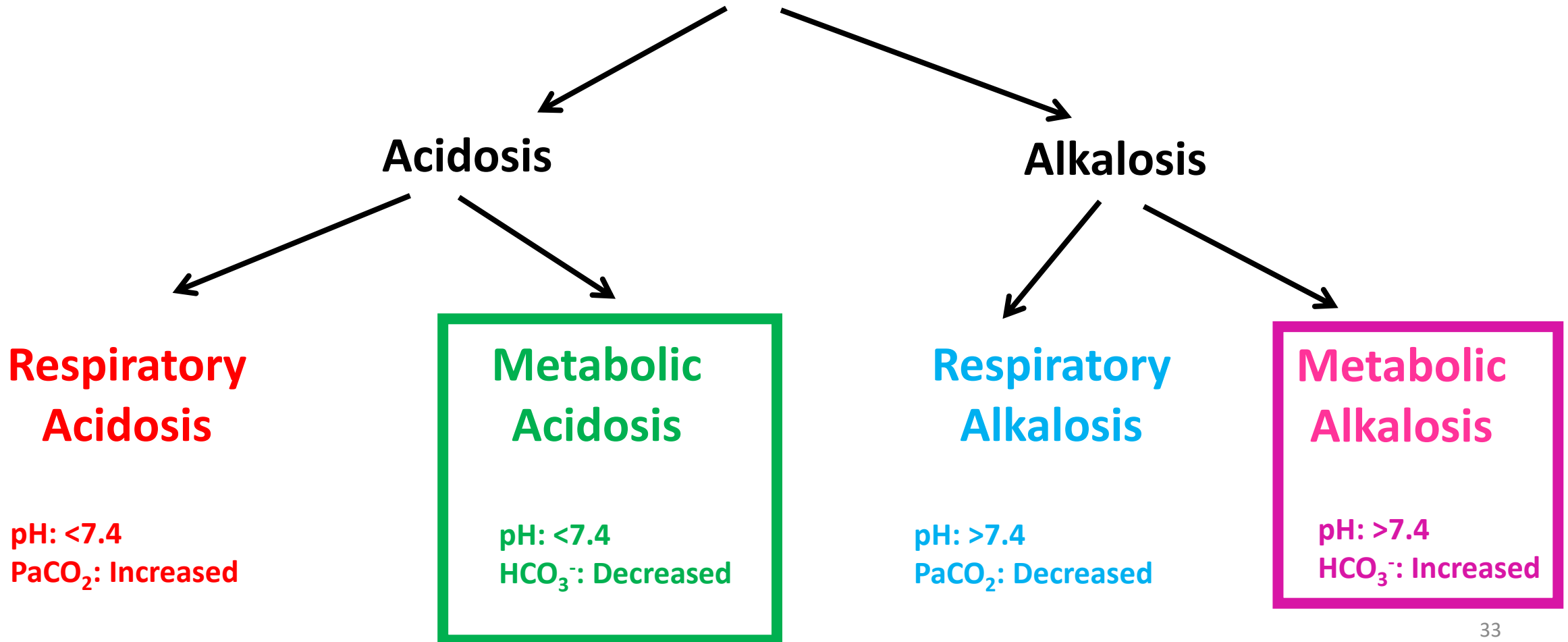


Summary: Mechanisms to maintain acid base homeostasis

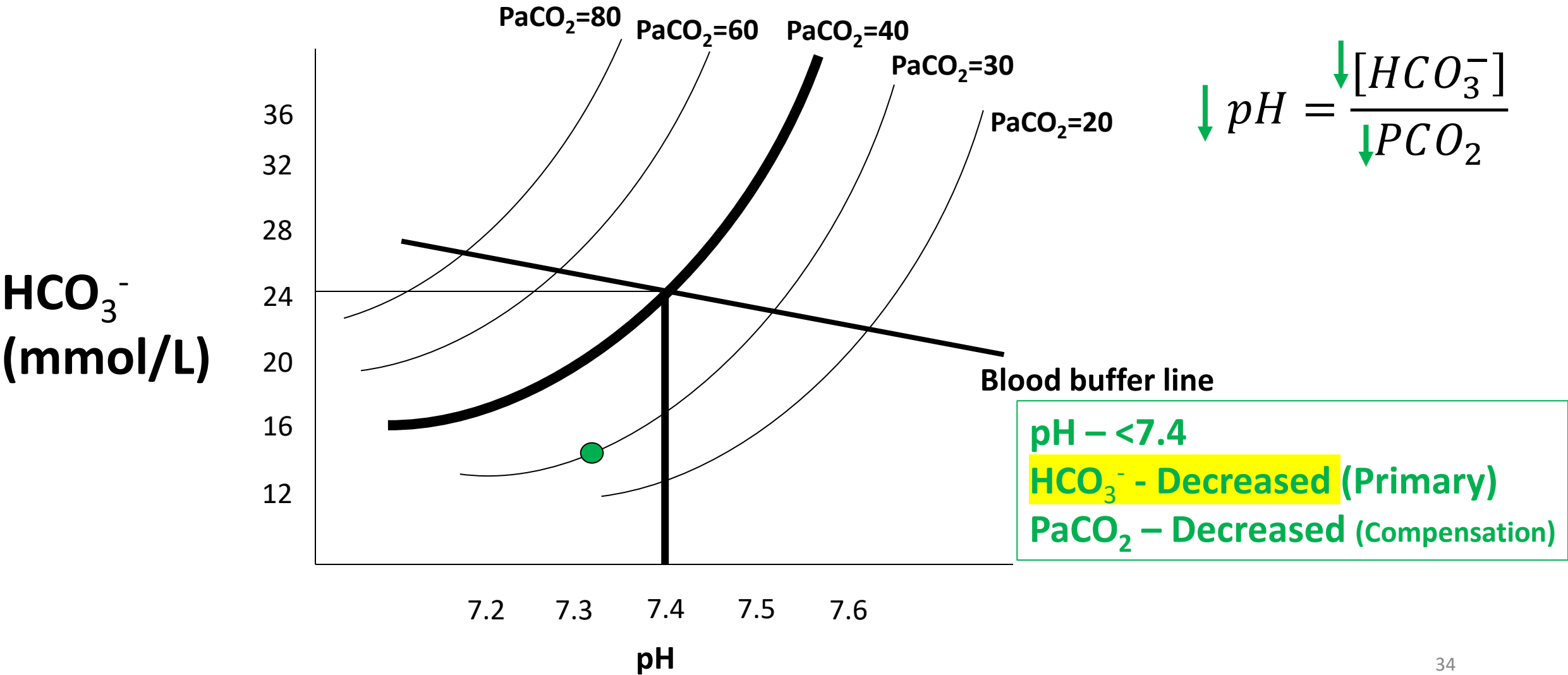
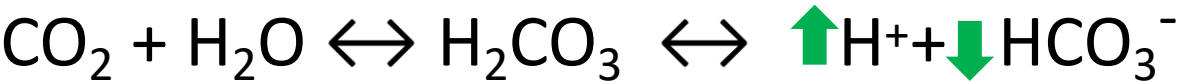
- Buffers – ECF and ICF buffers are the first line of defense
 - Bicarbonate (ECF)
 - Phosphate (ICF and urine)
 - Proteins (RBC and ICF)
 - Ammonia (urine)
- Respiratory system (responds in minutes)
 - Removal of CO_2
 - Alveolar ventilation regulates PaCO_2
- Renal system (responds in hours - days)
 - **Reabsorption** of HCO_3^-
 - **Maintenance of serum HCO_3^- levels** (form **NEW** HCO_3^-)
 - Excretion of H^+
 - Excretion of non-volatile/fixed acids (sulfate, phosphate, lactate)

Classification of Acid-base disorders

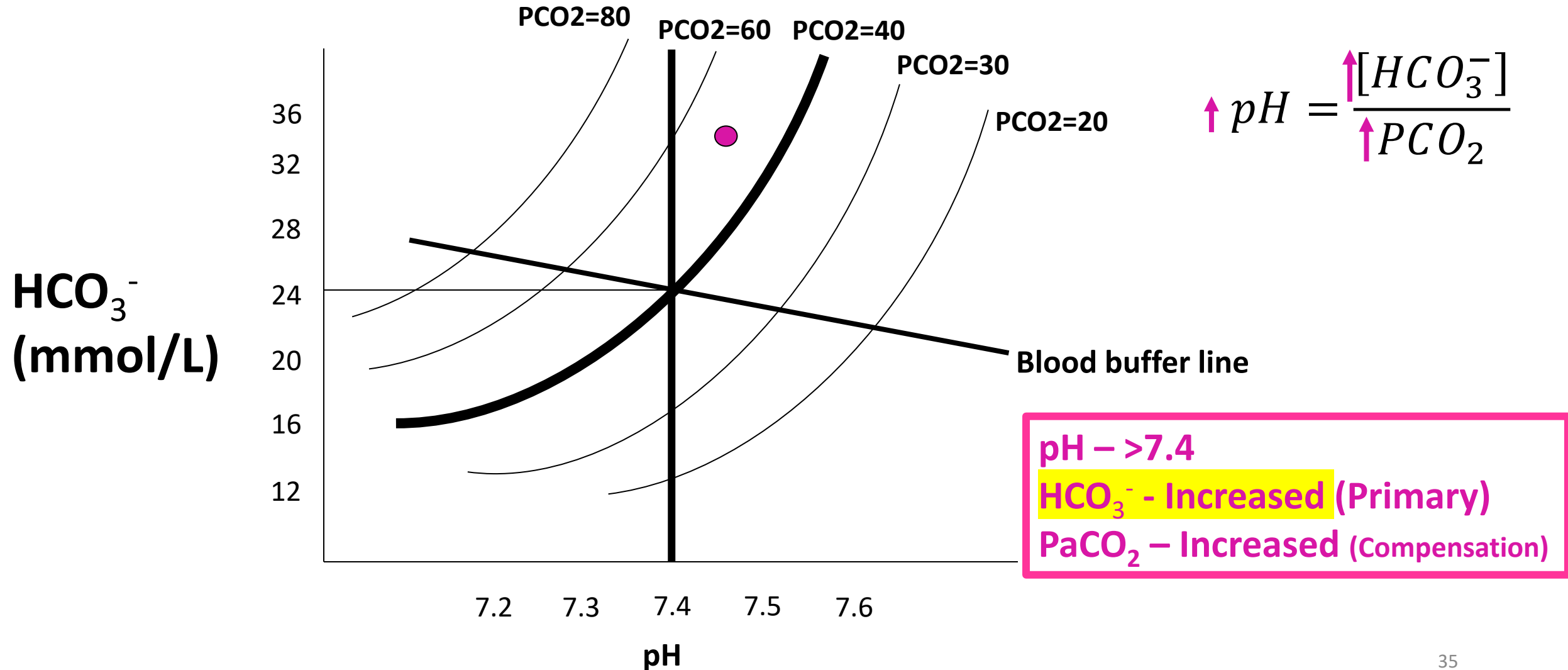
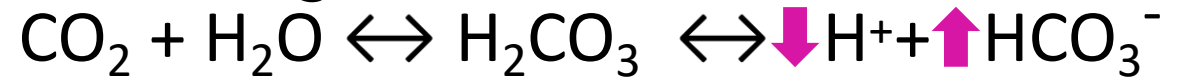
Based on pH



Metabolic acidosis ($\downarrow\downarrow\text{HCO}_3^-$)



Metabolic alkalosis ($\uparrow \uparrow \text{HCO}_3^-$)



Davenport Diagram

